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align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[subtitle] linebreak() text(fill: black70, size: 6.8pt)[Five DRIVE training images. Filtering uses the
green channel only. Noisy metrics average over 10 AWGN draws.] ] v(8pt) image-grid(data,
[Gradient magnitude, clean retinal images], "magnitude_path", show-input: true)
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align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[Orientation maps. Hue encodes edge angle modulo  $\pi$  and brightness follows normalized
magnitude.] ] v(8pt) image-grid(data, [Gradient orientation, clean retinal images],
"orientation_path", show-input: false)
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align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[Quantitative summary on vessel-boundary ODS, boundary-vector RMSE, and centerline tangent
orientation MAE.] ] v(8pt) grid( columns: 2, column-gutter: 10pt, [ metric-chart(data, "ods_f_score",
[Soft-label ODS F-score versus SNR], [ODS F-score]) ], [ metric-chart(data,
"gradient_vector_rmse_mean", [Gradient-vector RMSE versus SNR], [Vector RMSE]) ], [ metric-
chart(data, "orientation_mae_deg_mean", [Centerline tangent MAE versus SNR], [Angle MAE
(deg)]) ], [ summary-table(data) ], ) v(10pt) legend(data)

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align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[WVF radius-trace follow-up on the fixed DRIVE image set.] linebreak() text(fill: black70, size: 6.8pt)
[Garnet trace points sweep WVF across  $(r, d) = (3, 5), (5, 9), (9, 11), (15, 11), (25, 11), (50, 11)$ .
Thin horizontal lines are the fixed baseline methods. Outlined WVF points beat the best fixed
baseline at that metric and SNR. ] v(8pt) grid( columns: 2, column-gutter: 10pt, row-gutter: 10pt,
[ wvf-trace-chart(data, "ods_f_score", "inf", [ODS versus WVF radius, clean], [ODS F-score]) ], [ wvf-
trace-chart(data, "gradient_vector_rmse_mean", "inf", [Vector RMSE versus WVF radius, clean],
[Vector RMSE]) ], [ wvf-trace-chart(data, "ods_f_score", "20", [ODS versus WVF radius, 20 dB], [ODS
F-score]) ], [ wvf-trace-chart(data, "gradient_vector_rmse_mean", "20", [Vector RMSE versus WVF
radius, 20 dB], [Vector RMSE]) ], [ wvf-trace-chart(data, "ods_f_score", "10", [ODS versus WVF
radius, 10 dB], [ODS F-score]) ], [ wvf-trace-chart(data, "gradient_vector_rmse_mean", "10", [Vector
RMSE versus WVF radius, 10 dB], [Vector RMSE]) ], [ wvf-trace-chart(data, "ods_f_score", "5", [ODS
versus WVF radius, 5 dB], [ODS F-score]) ], [ wvf-trace-chart(data, "gradient_vector_rmse_mean",
"5", [Vector RMSE versus WVF radius, 5 dB], [Vector RMSE]) ], ) v(8pt) legend(data)

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align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[WVF centerline orientation trace on DRIVE.] ] v(8pt) grid( columns: 2, column-gutter: 10pt, row-
gutter: 10pt, [ wvf-trace-chart(data, "orientation_mae_deg_mean", "inf", [Orientation MAE versus
WVF radius, clean], [Angle MAE (deg)]) ], [ wvf-trace-chart(data, "orientation_mae_deg_mean",
"20", [Orientation MAE versus WVF radius, 20 dB], [Angle MAE (deg)]) ], [ wvf-trace-chart(data,
"orientation_mae_deg_mean", "10", [Orientation MAE versus WVF radius, 10 dB], [Angle MAE
(deg)]) ], [ wvf-trace-chart(data, "orientation_mae_deg_mean", "5", [Orientation MAE versus WVF
radius, 5 dB], [Angle MAE (deg)]) ], ) v(8pt) legend(data)

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align(center)[text(fill: black90, size: 11pt, weight: "bold") [title] linebreak() text(fill: black70, size: 8pt) [Fine-grained WVF radius and degree sweep on the fixed DRIVE image set.] linebreak() text(fill: black70, size: 6.8pt) [The heatmap shows 10 dB centerline orientation MAE across the feasible (r, d) grid. Crossed cells were skipped by the Section 7 conditioning gate. The outlined cell is the annotated optimum.]] v(8pt) wvf-grid-heatmap(data, [10 dB centerline orientation MAE across the feasible WVF grid]) v(10pt) align(left)[text(fill: black90, size: 7.2pt, weight: "bold") [Annotated optimum.] text(fill: black90, size: 7.2pt) [r=3, d=1] text(fill: black90, size: 7.2pt) [with orientation MAE 29.8694.] linebreak() text(fill: black90, size: 7.2pt, weight: "bold") [Driver assessment.] text(fill: black90, size: 7.2pt) [the orientation-MAE optimum ranges from r=3 to r=7 across the DRIVE SNR sweep, so vessel scale sets the baseline while the variance floor favors wider averaging under noise.]]